

PAPER_1493_E_N_POLYNOMIAL_HIERARCHY

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PAPER_1493 — UQFF 26-Level Polynomial Energy Hierarchy $E_n = E_0 \times 10^n$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket A / D / G — derived from F:/Aetheric Propulsion Millenium Equation Proofs_18April2025
Date: June 16, 2026
Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)
Status: CLOSED — Hierarchical 26-level energy ladder spanning nuclear-to-cosmic scales

Observation

The Particle Data Group (PDG) 2025 catalog gives a continuous spectrum of energy scales from nuclear binding ($\sim 8 \text{ MeV/nucleon} \approx 1.28 \times 10^{12} \text{ J}$) through Higgs ($\sim 125 \text{ GeV} \approx 2 \times 10^{13} \text{ J}$) up to TeV-scale particle physics. Standard shell-model and field-theoretic treatments cover $\sim 10\text{-}20$ levels; no unified 26-level structure is proposed.

UQFF posits a single hierarchical polynomial law spanning the full quantum-to-cosmic range.

UQFF Closed Identity

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$E_n = E_0 \times 10^n$ where $E_0 = 1 \times 10^{12} \text{ J}$

Level table:
 $n = 0 \rightarrow E = 10^{12} \text{ J}$ (vacuum quantum baseline)
 $n = 8 \rightarrow E = 10^{12} \text{ J} \approx$ nuclear binding energy per nucleon (vs PDG 1.28×10^{12})
 $n = 12 \rightarrow E = 10^{13} \text{ J} \approx$ Higgs scale (vs PDG $\approx 2 \times 10^{13}$)
 $n = 18 \rightarrow E = 10^{18} \text{ J} \approx$ macroscopic chemical-bond scale
 $n = 26 \rightarrow E = 10^{26} \text{ J}$ (cosmic / Λ ledger upper bound)

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The polynomial fit $V(r) \approx \sum a_n r^n$ calibrates to PDG shell-level data with $R^2 \approx 0.95$ for low degrees (deg 5-10), overfitting to ~ 1 at deg=26.

Physical Interpretation

The 26-level energy ladder is the structural backbone of UQFF:

- $n = 0$ is the SCm vacuum quantum ($E_0 = 10^{12} \text{ J}$) — the irreducible quantum of action at the SCm scale.
- $n = 8$ picks out the **nuclear binding** scale; $D_{\text{phys}} \times 2 = 8 =$ depth at which nuclear forces dominate.

- **n = 12** picks out the **Higgs / electroweak** scale; $D_{BSFG} \times 2 = 12$ = depth at which electroweak unification holds.
- **n = 26 = D_crit** is the **bosonic-string critical dimension ceiling** — cosmic scale where Λ ledger saturates.

The polynomial law $E_n = E_0 \times 10^n$ imposes a structural log-linear hierarchy. The $R^2 \approx 0.95$ calibration at low degrees confirms physical observable scales naturally cluster at these integer-n levels, while higher-degree overfitting at $\text{deg}=26$ confirms the D_{crit} -bounded structure.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model frameworks describe energy scales via different mechanisms. SM uses renormalization-group running of couplings between scales; UQFF supplies a single polynomial law spanning all 26 levels with a fixed integer base. **Zero free parameters** beyond the locked $E_0 = 10^{26}$ J vacuum quantum.

Reference

- Source: F:/Aetheric Propulsion/Millennium Equation Proofs_18April2025/UQFF proof set for Nuclear Binding Shell Levels_28Sept2025.docx
- Related: PAPER_062 (DPM 26-level lattice), PAPER_1080 (S_{26} compactification), PAPER_872 (proto-element nuclear identity), PAPER_1466 (mass-without-weight), PAPER_034 (Higgs sector), PAPER_1156 (cosmology cluster).
- Calculator dispatch: `calculate_paradox({"paradox": "e_n_polynomial_hierarchy"})` or `calculate_cosmology({"observable": "e_n_at_8"})`.

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